

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 1

MARCH 2012

Mathematics Extension 2

General Instructions

- Working time - 70 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Start each question on a new page

Total marks - 52

- Attempt Questions 1 – 4
- All questions are of equal value

Name : _____

Teacher : _____

Question 1	Question 2	Question 3	Question 4	Total

Question 1 (13 marks)

- a) For the ellipse with equation $x^2 + 4y^2 = 16$ find
- i) the eccentricity 2
 - ii) the coordinates of the foci 1
 - iii) the equation of the directrices 1
 - iv) the length of the chord of the ellipse which passes through the focus and is perpendicular to the major axis of the ellipse 2
- b) Factorise $x^2 + 6x + 25$ over the complex field 2
- c) i) Express $1 + i\sqrt{3}$ in modulus argument form. 2
- ii) Find the smallest positive integer value of n such that 3

$$\operatorname{Im} \left(\frac{-1+i}{1+i\sqrt{3}} \right)^n = 0$$

Question 2 (13 marks) - Start a new page

a) If $z = 2 + i$ and $w = 3 - 2i$ find simplified expressions for

i) $z + \bar{w}$ 1

ii) $\frac{z}{w}$ 2

b) Sketch the locus of z described by the following -

i) $0 \leq \text{Arg}(z - 2i) \leq \frac{\pi}{6}$ 2

ii) $\text{Im}(z^2) = |z - \bar{z}|$ 3

c) Solve $z^2 = 7 + i\sqrt{72}$ over the complex field, 2

giving your answer in the form $x + iy$ where x and y are real.

d) Given $\cos(x - y) = y \cos x$ 3

show that $\frac{dy}{dx} = \frac{\sin(x-y) - y \sin x}{\sin(x-y) - \cos x}$

Question 3 (13 marks) - Start a new page

a) $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$.

i) Show that the equation of the normal to the above ellipse at the point P 3

is given by the equation $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$

ii) The normal found in part i) meets the major axis of the ellipse at the point G.

If S is a focus of the ellipse and e its eccentricity, 4

show that $SG = eSP$

b) Find all the solutions of $z^6 = -1$ 3

c) Simplify $\frac{(\sin \frac{\pi}{5} + i \cos \frac{\pi}{5})^2}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$ 3

Question 4 (13 marks) - Start a new page

- a) i) Draw a neat sketch of the locus represented by 2

$$|z + \sqrt{2} - i\sqrt{2}| = 1$$

- ii) For z on the locus in part i) find

$\alpha)$ the minimum value of $|z|$ 1

$\beta)$ the minimum value of $\text{Arg}(z)$ 1

- b) z is a complex number such that $\text{Arg}(z) = \theta$ where $\frac{\pi}{4} < \theta < \frac{\pi}{2}$.

Find expressions for the following in terms of θ ,

i) $\text{Arg}(iz + z)$ 2

ii) $\text{Arg}(iz - z)$ 2

- c) For the general ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b$ describe the 1

effect on the ellipse as $e \rightarrow 0$ (e is the eccentricity)

- d) Show by Mathematical Induction that 4

$$(1 - a_1)(1 - a_2) \dots (1 - a_n) > 1 - (a_1 + a_2 + \dots + a_n)$$

for all positive integers n where $n > 1$, if a_k satisfies $0 < a_k < 1$ for $1 \leq k \leq n$.

SOLUTIONS

1

a) 1) $x^2 + 4y^2 = 16$

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$b^2 = a^2(1 - e^2)$$

$$4 = 16(1 - e^2)$$

$$\frac{1}{4} = 1 - e^2$$

$$e^2 = \frac{3}{4}$$

$$e = \frac{\sqrt{3}}{2}$$

ii) $S(\pm ae, 0)$

$$S(\pm 2\sqrt{3}, 0)$$

iii) $x = \pm \frac{a}{e}$

$$x = \pm \frac{8}{\sqrt{3}}$$

iv) when $x = 2\sqrt{3}$

$$12 + 4y^2 = 16$$

$$4y^2 = 4$$

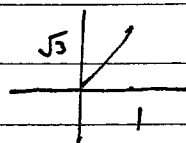
$$y^2 = 1$$

$$y = \pm 1$$

$\therefore$ length = 2 units

$$\begin{aligned}
 b) \quad & x^2 + 6x + 25 \\
 & x^2 + 6x + 9 + 16 \\
 & (x+3)^2 + 4^2 \\
 & (x+3+4i)(x+3-4i)
 \end{aligned}$$

$$c) \quad i) \quad 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$



$$ii) \quad -1+i = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$\frac{-1+i}{1+i\sqrt{5}} = \frac{\sqrt{2} \operatorname{Cis} \frac{3\pi}{4}}{2 \operatorname{Cis} \frac{\pi}{5}}$$

$$= \frac{1}{\sqrt{2}} \operatorname{Cis} \frac{5\pi}{12}$$

$$\text{for } \operatorname{Im} \left(\frac{-1+i}{1+i\sqrt{5}} \right)^n = 0$$

$$n \times \frac{5\pi}{12} = m\pi \quad \text{where } m \text{ is an integer}$$

$$\therefore n = 12$$

$$c) \quad z^2 = 7 + i\sqrt{72}$$

$$(x+iy)^2 = 7 + i\sqrt{72}$$

$$x^2 - y^2 + 2ixy = 7 + i\sqrt{72}$$

$$x^2 - y^2 = 7 \quad 2xy = \sqrt{72}$$

$$xy = 3\sqrt{2}$$

$$\therefore x=3, y=\sqrt{2}$$

$$\therefore z = 3 + i\sqrt{2}, -3 - i\sqrt{2}$$

$$d) \quad \cos(x-y) = y \cos x$$

$$-\sin(x-y) \left(1 - \frac{dy}{dx}\right) = \frac{dy}{dx} \cos x + y \sin x$$

$$-\sin(x-y) + \sin(x-y) \frac{dy}{dx} = \frac{dy}{dx} \cos x + y \sin x$$

$$y \sin x - \sin(x-y) = \frac{dy}{dx} (\cos x - \sin(x-y))$$

$$\frac{dy}{dx} = \frac{y \sin x - \sin(x-y)}{\cos x - \sin(x-y)}$$

[2]

a) i) $z + \bar{w}$

$$= 2+i + 3+2i$$

$$= 5+3i$$

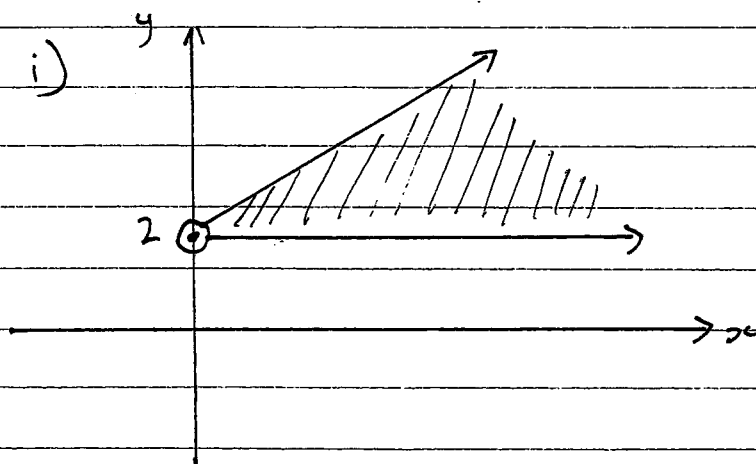
ii) $\frac{z}{w} = \frac{2+i}{3-2i}$

$$= \frac{2+i}{3-2i} \times \frac{3+2i}{3+2i}$$

$$= \frac{6+4i+3i-2}{9+4}$$

$$= \frac{4+7i}{13}$$

b) i)



ii) $\text{Im}(z^2) = |z - \bar{z}|$

$$\text{Im}(x^2 - y^2 + 2ixy) = |x+iy - (x-iy)|$$

$$2xy = |2y|$$

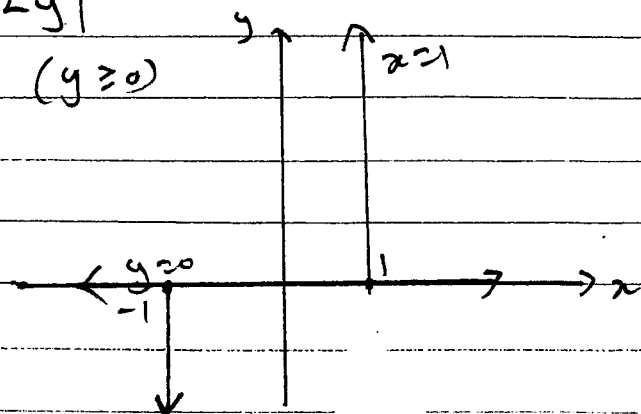
$$xy + y = 0 \quad (y \leq 0)$$

$$y = 0, x = -1$$

$$xy - y = 0 \quad (y \geq 0)$$

$$y(x-1) = 0$$

$$y = 0, x = 1$$



$$\boxed{3} \quad a) \quad i) \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$

$$\therefore m_T = \frac{-b^2 a \cos \theta}{a^2 b \sin \theta}$$

$$= -\frac{b \cos \theta}{a \sin \theta}$$

$$\therefore m_N = \frac{a \sin \theta}{b \cos \theta}$$

using $y - y_1 = m(x - x_1)$

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$by \cos \theta - b^2 \sin \theta \cos \theta = ax \sin \theta - a^2 \sin \theta \cos \theta$$

$$ax \sin \theta - by \cos \theta = a^2 \sin \theta \cos \theta - b^2 \sin \theta \cos \theta$$

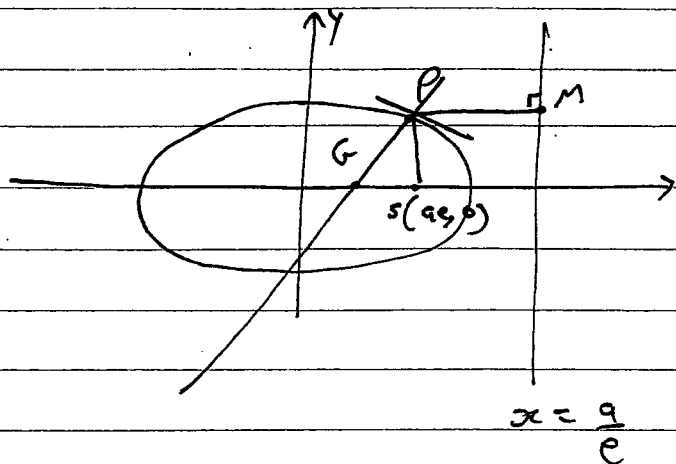
$$\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$$

11) when $y=0$

$$\frac{ax}{c \cos \theta} = a^2 - b^2$$

$$x = \frac{(a^2 - b^2) \cos \theta}{a}$$

$$\therefore G \left(\frac{(a^2 - b^2) \cos \theta}{a}, 0 \right)$$



$$\therefore e \cdot SP = ex \cdot e \cdot PM$$

$$= e^2 \left(\frac{a}{e} - a \cos \theta \right)$$

$$= e (a - ae \cos \theta)$$

$$SG = ae - \left(\frac{a^2 - b^2}{a} \cos \theta \right)$$

$$= ae - \left(\frac{a^2 e^2}{a} \right) \cos \theta$$

$$= ae - ae^2 \cos \theta$$

$$= e (a - ae \cos \theta)$$

$$= e \cdot SP$$

$$b^2 = a^2 (1 - e^2)$$

$$\frac{b^2}{a^2} = 1 - e^2$$

$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$a^2 e^2 = a^2 - b^2$$

$$b) \quad z^6 = -1$$

$$z_1 = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$$

$$z_2 = i$$

$$z_3 = \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}$$

$$z_4 = \cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6}$$

$$z_5 = -i$$

$$z_6 = \cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}$$

$$c) \quad \frac{(\sin \frac{\pi}{5} + i \cos \frac{\pi}{5})^2}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$$

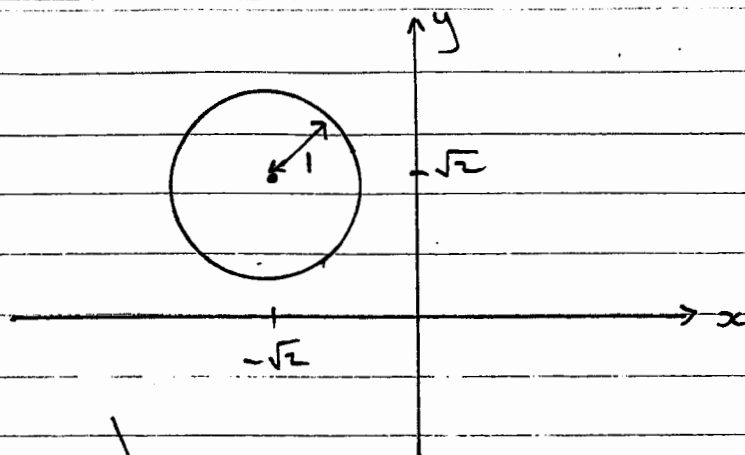
$$= \frac{(\cos(\frac{\pi}{2} - \frac{\pi}{5}) + i \sin(\frac{\pi}{2} - \frac{\pi}{5}))^2}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$$

$$= \frac{(\cos \frac{3\pi}{10} + i \sin \frac{3\pi}{10})^2}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$$

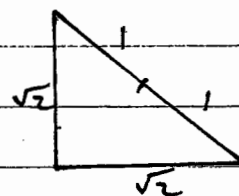
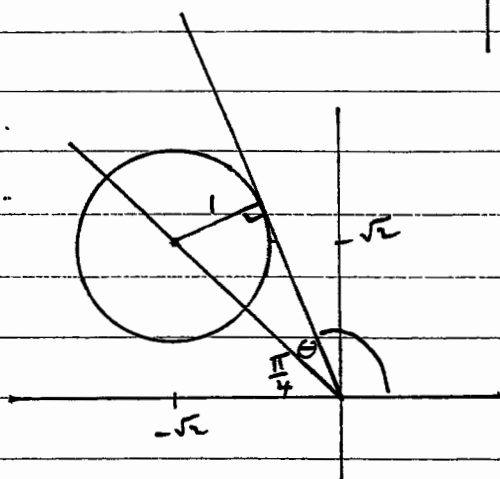
$$= \frac{\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}}{\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}}$$

$$= \cos \frac{13\pi}{20} + i \sin \frac{13\pi}{20}$$

4) a) i)



ii)



$$\alpha) \min |z| = 1$$

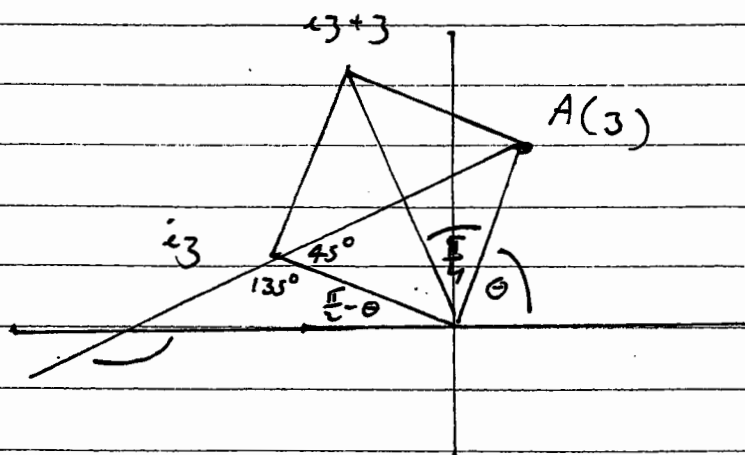
$$\beta) \min \operatorname{Arg}(z) = \frac{\pi}{2} + \frac{\pi}{12} = \frac{7\pi}{12}$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}$$

$$\frac{\pi}{2} - \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12}$$

b)



$$i) \operatorname{Arg}(i3+3) = \frac{\pi}{4} + \theta$$

$$ii) \operatorname{Arg}(23-3) = -\left(\frac{3\pi}{4} + \frac{\pi}{2} - \theta\right) = \theta - \frac{5\pi}{4}$$

c) the ellipse tends towards a circle.

d) Step 1 test $n=2$.

$$\begin{aligned} \text{LHS} &= (1-a_1)(1-a_2) \\ &= 1-a_1-a_2+a_1a_2 \\ &> 1-a_1-a_2 \\ &= \text{RHS} \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 1-(a_1+a_2) \\ &= 1-a_1-a_2 \end{aligned}$$

$\therefore$ true for $n=2$.

Step 2 ... assume true for $n=k$

$$\text{i.e. } (1-a_1)(1-a_2)\dots(1-a_k) > 1-(a_1+a_2+\dots+a_k)$$

show true for $n=k+1$

$$(1-a_1)(1-a_2)\dots(1-a_k)(1-a_{k+1})$$

$$> [1-(a_1+a_2+\dots+a_k)](1-a_{k+1}) \quad \text{from assumption}$$

$$= 1-(a_1+a_2+\dots+a_k)-a_{k+1}+(a_1+a_2+\dots+a_k)a_{k+1}$$

$$> 1-(a_1+a_2+\dots+a_k)-a_{k+1}$$

$$= 1-(a_1+a_2+\dots+a_k+a_{k+1})$$

which is the required result

$\therefore$ true for $n=k+1$ if true for $n=k$.

Step 3

As true for $n=2$, also true for $n=2+1$, i.e. $n=3$

As true for $n=3$, also true for $n=3+1$, i.e. $n=4$

and so on for all positive integers n , $n > 1$.